VISHAL YADAV

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INTRODUCTION

AI researcher, enterprise AI product builder, and founder with eight years at the frontier of applied machine learning, agentic reasoning, and structured knowledge extraction. Currently leading enterprise AI governance transformation and structured-retrieval LLM research (ASIMOV) at BlackRock AI Labs, while co-building DigiSam Foundation to close the AI divide for rural women in India. Proven track record of taking provenance-grounded multi-agent systems and unstructured-to-knowledge-graph engines from 0-to-1 into production across global finance (BlackRock, Barclays), government (Dubai Dept. of Finance), and UK Defence. Deeply focused on translating calibrated, evidence-traced AI reasoning and closed-loop active learning into predictable drug discovery and translational therapeutics, supported by independent PoC design in multistate conditional biologics and wet-lab research links at the University of Cambridge.

EXPERIENCE

Senior Data Scientist
BlackRock AI Labs

July 2025 – Present Edinburgh, United Kingdom

- ***AI Governance for Vision 2030:*** Leading a high-profile cross-functional data science team to reimagine BlackRock's enterprise AI governance. Translated complex policies into **automated decision logic** that scans codebases and assesses application risk in real time, making governance an innovation enabler rather than a compliance bottleneck.
- ***ASIMOV – AI-Powered Investment Research:*** Leading the agentic data visualisation and structured retrieval workstream, delivering **text-to-SQL** and **SQL-to-chart** capabilities at scale across financial filings. Research focus: improving provenance-grounded structured data retrieval. Featured in *Bloomberg* and the *Wall Street Journal*.

Co-Founder & Director
DigiSam Foundation

May 2025 – Present Uttar Pradesh, India

- ***0-to-1 Venture Strategy & Partnerships:*** Lead organisational strategy, external partnerships, governance, and programme design across three workstreams: **AI literacy training**, **last-mile welfare delivery**, and **accessible AI tooling** for low-infrastructure communities.
- ***Evidence-Led Field Execution (Survey Sakhi):*** Designed and launched a census-level baseline survey across all **250 households** in Saidbhar village, UP. Key finding: 60% own smartphones yet 87% have never accessed a government service app; only 1.6% had heard of AI, yet 31% requested AI training. Translated field data into **167 women (67%)** enrolling in digital literacy and government-certified AI programmes.

Senior Consultant – Data Science & GenAI
Capgemini Invent

Dec 2021 – July 2025 Glasgow, United Kingdom

- ***Project ASCEND (Dubai Dept. of Finance):*** Originated and built a multi-agent reasoning system performing automated gap analysis of new financial processes against existing regulations, generating evidence-traced recommendations and KPIs and reducing report turnaround from weeks to minutes (**Azure, LlamaIndex, Python, Next.js**). Covered by Gulf News, Finance Middle East, and the DOF.
- ***Knowledge Graphs from Unstructured Data (UK Defence):*** Architected LLM pipelines to extract structured, provenance-linked knowledge graphs from multi-format unstructured intelligence at scale, enabling a digital model of UK defence and cutting modelling time from weeks to minutes (**Azure, OpenAI, Python**).

Data Scientist
Barclays PLC

July 2018 – Dec 2021 Glasgow, United Kingdom

- Built and deployed an equity trade volume forecaster for Barclays Smart Investor on **AWS** (EMR, S3, SageMaker) using **Keras, TensorFlow, and Spark**, complete with a full-stack **Angular** KPI dashboard.

Assistant Professor
Meerut Institute of Engineering and Technology

Jan 2018 – May 2018 Meerut, India

- Taught **Machine Learning and Pattern Recognition** to undergraduate students; organised a face detection hackathon and mentored a student team to the finals of the **Smart India Hackathon**.

TRANSLATIONAL BIO-AI & R&D

AlloSwitch – Closed-Loop Active Learning for Conditional Biologics
Independent PoC

- ***Multistate Negative Design & Active-Learning Architecture:*** Working on a technical alpha for AI-designed, target-binding-triggered conformational protein switches. Formulated state-function schemas linking sequence and fusion geometry to measured leak, activation ratio, and reversibility across OFF/ON conformational ensembles (evaluating **RFdiffusion, ProteinMPNN, and BioEmu** limits).
- ***Calibrated Small-Data Modelling & Wet-Lab Integration:*** Designing strict gate-based experimental validation protocols (constitutive-ON, locked-OFF, manual non-AI baseline, and orthogonal luminescence + **BLI/SPR** readouts) paired with data-efficient **Bayesian optimisation** and **Gaussian-process active learning** for 24–48 construct libraries, informed by translational nanocarrier delivery research at the **University of Cambridge**.

EDUCATION

MSc in Machine Learning, Data Mining and High-Performance Computing
University of Bristol

Aug 2016 – July 2017

Thesis: Synthetic faces for intelligent personal assistants

B.Tech. in Computer Science
Dr. A.P.J. Abdul Kalam University

July 2011 – May 2015

PURSUITS & AWARDS

MasterChef UK Contestant 2025

Selected from tens of thousands of applicants nationwide, competing on BBC national television and executing complex dishes under intense time pressure before Britain's toughest culinary critics.

Google GenAI Global Hackathon Finalist July 2023

Built a PoC to analyse and surface media bias in Spotify news podcasts using Google's GenAI stack.

Expedia & ENTER2020 Hackathon Jan 2020

Won Best Data Visualisation Award.

Endurance & Community Impact 2017 – Present

2 full marathons, 3 half marathons; co-organised three community running events; Volunteer at Save the Children (2020–2024).

SKILLS

AI, Reasoning & Evaluation

LLMs | Agentic Systems | Knowledge Graphs | Structured Retrieval (Text-to-SQL) | Active Learning | Benchmark & Contamination Design

Engineering & Cloud

Python | SQL | AWS | Azure | PostgreSQL (PGVector) | Spark | Next.js

Venture & Leadership

0-to-1 Venture Strategy | Enterprise AI Governance | C-Suite & Regulatory Stakeholder Management | Cross-Functional Team Building